Notes: Barber, Lehavy, McNichols, and Trueman (JF, 2001)

Can Investors Profit from the Prophets? Security Analyst Recommendations and Stock Returns

Contents

1 Big picture

- **Research question.** Can investors earn abnormal returns by trading on publicly available security analyst recommendations? More specifically, do consensus analyst ratings contain information that is not immediately incorporated into prices?
- **Main idea.** Analysts spend resources collecting and interpreting firm information. If their consensus recommendations are informative and prices adjust slowly, then stocks with more favorable recommendations should outperform stocks with less favorable recommendations.
- **Empirical object.** The paper forms daily value-weighted portfolios based on each stock's consensus analyst recommendation and evaluates raw, market-adjusted, and risk-adjusted returns in calendar time.
- **Main gross-return finding.** Buying the most favorably rated stocks and shorting the least favorably rated stocks earns economically large abnormal gross returns when portfolios are rebalanced daily and investors react quickly to recommendation changes.
- **Main trading-cost finding.** The strategies require very high turnover. Once realistic transaction costs are applied, abnormal net returns are not reliably greater than zero.
- **Important asymmetry.** Delays and less frequent rebalancing reduce the returns to buying the most favorably recommended stocks much more than they reduce the negative abnormal returns of the least favorably rated stocks.
- **Size result.** Predictability is strongest among small firms. Among the few hundred largest firms, the return spread between highly and poorly rated stocks is not reliably different from zero.
- **Economic takeaway.** Analyst recommendations appear informative, but the costs and speed needed to exploit them make the documented strategies hard to convert into reliably positive net trading profits.

2 Incremental contribution of the paper

- **Investor-oriented perspective.** Earlier studies such as Stickel and Womack study event-time price reactions to individual recommendation changes. This paper asks the more practical question: what happens to an investor who trades portfolios based on consensus recommendations in real calendar time?
- **Consensus recommendations.** The paper focuses on average analyst ratings, which are easy for investors to access and summarize the information from all analysts covering a firm.
- **Large recommendation database.** The sample uses the Zacks database from 1985 to 1996, covering 361,620 recommendations after matching to CRSP.
- **Daily portfolio updating.** The portfolio methodology updates consensus ratings whenever coverage is initiated, changed, or dropped, and rebalances at the close of trading on the day of the change.
- **Transaction-cost discipline.** The paper goes beyond abnormal gross returns by estimating turnover and translating turnover into trading costs.

- **Delay and rebalancing tests.** The paper separately studies slower rebalancing and delayed reaction to recommendations, making clear how quickly any trading signal decays.
- **Size partition.** The analysis documents that the evidence is concentrated in small firms, where trading costs and implementation frictions are also largest.

3 Data and measurement

3.1 Analyst recommendation data

The main recommendation data come from Zacks Investment Research. Zacks collects analyst ratings from written and electronic brokerage-house reports.

- **Sample period.** Recommendations cover 1985 through 1996.
- **Recommendation scale.** Ratings run from 1 to 5: 1 is strong buy, 2 is buy, 3 is hold, 4 is sell, and 5 is strong sell. A rating of 6 denotes termination of coverage.
- **Final sample.** The raw Zacks database contains 378,326 observations. After dropping firms not on CRSP, the final sample contains 361,620 recommendations.
- **Broker and analyst coverage.** The data include recommendations from 269 brokerage houses and 4,340 analysts over the sample.
- **Database caveat.** The academic Zacks database omits some brokerage-house recommendations because of distribution agreements. The paper notes that Zacks consensus statistics may include recommendations not distributed in the academic database.

3.2 Consensus recommendation measure

For firm i on date $\tau - 1$, the consensus recommendation is the average of the outstanding individual analyst recommendations:

$$A_{i,\tau-1} = \frac{1}{n_{i,\tau-1}} \sum_{j=1}^{n_{i,\tau-1}} A_{ij,\tau-1}. \tag{1}$$

Lower values are more favorable because rating 1 is a strong buy.

3.3 Portfolio groups

At the close of each trading day, every covered stock is assigned to one of five portfolios:

- **Portfolio 1:** $1 \leq A \leq 1.5$, the most favorable consensus recommendations.
- **Portfolio 2:** $1.5 < A \leq 2$.
- **Portfolio 3:** $2 < A \leq 2.5$.
- **Portfolio 4:** $2.5 < A \leq 3$.

- **Portfolio 5:** $A > 3$, the least favorable consensus recommendations.

The paper also forms an **all covered** portfolio, a **neglected** portfolio of CRSP firms without Zacks coverage, and a long-short covered-minus-neglected portfolio.

3.4 Returns, turnover, and trading costs

- **Value weighting.** Portfolios are value-weighted rather than equal-weighted. This avoids overstating returns because of bid-ask bounce and better reflects economic significance.
- **Monthly returns.** Daily portfolio returns are compounded to obtain monthly returns.
- **Turnover.** Daily turnover is the fraction of the previous day's portfolio that must be sold after recommendation changes and rebalancing. Annual turnover multiplies the daily average by the number of trading days.
- **Transaction costs.** The paper uses estimates from Keim and Madhavan to approximate round-trip transaction costs: 0.727% for large firms, 1.94% for medium firms, and 4.12% for small firms. A representative portfolio-level round-trip cost is estimated at 1.31% of value traded.

4 Methodology and empirical specifications

4.1 Baseline calendar-time portfolio test

After forming each portfolio as of the close of date $\tau - 1$, the value-weighted return on portfolio p on date τ is

$$R_{p\tau} = \sum_{i=1}^{n_{p,\tau-1}} X_{i,\tau-1} R_{i\tau}, \tag{2}$$

where $X_{i,\tau-1}$ is firm i's share of the market capitalization of portfolio p at the close of date $\tau - 1$.

Monthly returns are compounded from daily returns:

$$R_{pt} = \prod_{\tau=1}^{n} (1 + R_{p\tau}) - 1. \tag{3}$$

4.2 Abnormal return models

The paper evaluates abnormal performance using three models.

CAPM:

$$R_{pt} - R_{ft} = \alpha_p + \beta_p (R_{mt} - R_{ft}) + \varepsilon_{pt}. \tag{4}$$

Fama-French three-factor model:

$$R_{pt} - R_{ft} = \alpha_p + \beta_p (R_{mt} - R_{ft}) + s_p SMB_t + h_p HML_t + \varepsilon_{pt}. \tag{5}$$

Four-characteristic model with momentum:

$$R_{pt} - R_{ft} = \alpha_p + \beta_p(R_{mt} - R_{ft}) + s_p SMB_t + h_p HML_t + m_p PMOM_t + \varepsilon_{pt}. \tag{6}$$

The intercept α_p is interpreted as abnormal gross return before transaction costs.

4.3 Delayed reaction and less frequent rebalancing

The paper studies two implementation frictions.

- **Less frequent rebalancing.** Instead of rebalancing daily, portfolios are rebalanced weekly, semimonthly, or monthly.
- **Delayed reaction.** Investors are assumed to react to consensus recommendation changes after one week, half a month, or one month, while still rebalancing daily once the delayed information is used.

These tests separate two practical issues: how often investors need to trade and how quickly they need to observe recommendation changes.

4.4 Size partitions

The paper divides stocks into small, medium, and large groups using NYSE market-capitalization cutoffs. Small firms are in the bottom 30% of NYSE market capitalization, large firms are in the top 30%, and medium firms are in between. The paper then re-estimates the recommendation portfolios within each size group.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether publicly available consensus analyst recommendations forecast abnormal returns in implementable calendar-time portfolios. The evidence is strongest for the existence of return predictability before trading costs.

5.2 What the paper cannot fully identify

The paper does not randomly assign analyst coverage or recommendations. It therefore cannot cleanly separate analyst skill, investor underreaction, correlated omitted signals, and endogenous coverage choices. Its contribution is primarily to measure the returns and costs of trading on the signal investors can observe.

5.3 Why end-of-day rebalancing matters

The baseline strategy assumes that investors trade at the close on the day a recommendation change occurs. This excludes any return from advance knowledge or intraday reaction before the recommendation becomes public. Even with that conservative timing, the gross return spread is large.

5.4 Why net returns matter

The gross-return evidence alone might suggest an exploitable anomaly. The turnover evidence changes the interpretation. Daily rebalancing produces annual turnover often above 400%, so even modest round-trip trading costs can eliminate the abnormal returns.

5.5 Economic interpretation of the asymmetry

The least favorably rated stocks continue to underperform even when investors react with a delay. This is consistent with negative recommendation information drifting into prices more slowly than positive recommendation information, or with downgrades conveying information that persists longer.

5.6 Practical investor implication

The paper does not imply that recommendations are useless. Even if active trading strategies do not reliably earn positive net abnormal returns, investors already planning to trade may still benefit from preferring more favorably rated stocks and avoiding or selling less favorably rated stocks.

6 Tables and figures

6.1 Figure 1: annualized gross returns by recommendation portfolio

Read:

- Figure 1 reports annualized geometric mean gross returns from 1986 to 1996.
- Portfolio 1, the most favorable recommendation portfolio, earns 18.8% annually.
- Portfolio 5, the least favorable recommendation portfolio, earns only 5.8% annually.
- The market benchmark earns 14.5% annually.
- Raw performance decreases monotonically as recommendations become less favorable.

Economic message: There is a large raw-return spread between highly recommended and poorly recommended stocks. This motivates the formal risk-adjusted and trading-cost analysis.

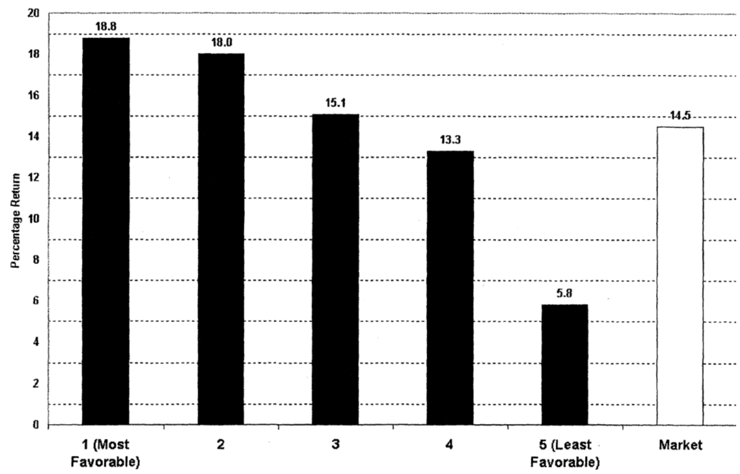

Figure 1. Annualized geometric mean percentage gross return earned by portfolio formed on the basis of consensus analyst recommendations, 1986 to 1996.

Figure 1: Annualized geometric mean percentage gross returns by consensus recommendation portfolio.

6.2 Tables I and II: recommendation data and transition patterns

Read Table I:

- Table I reports descriptive statistics for the Zacks recommendation database.
- Zacks coverage expands from 1,841 covered firms in 1985 to 5,628 covered firms in 1996.
- Covered firms represent 27.0% of listed firms in 1985 and 59.8% in 1996.
- By market capitalization, coverage is much broader: covered firms represent 68.8% of market capitalization in 1985 and 95.6% in 1996.
- The average rating declines from 2.52 in 1985 to 2.04 in 1996, meaning recommendations become more favorable over time.

Read Table II:

- Table II is a transition matrix for individual analyst recommendations.
- Most observations are concentrated in ratings 1, 2, and 3; sell and strong-sell recommendations are rare.
- Of all non-dropped recommendations, 29.8% are rating 1, 24.3% are rating 2, 39.5% are rating 3, and only 6.5% are ratings 4 or 5.
- First Zacks recommendations are usually buys or holds, again showing analysts' reluctance to initiate with sell ratings.

Economic message: The analyst recommendation environment is optimistic and coverage expands substantially over time. Because outright sell recommendations are rare, the least favorable consensus portfolio is small and potentially especially informative.

Table I

Descriptive Statistics on Analyst Recommendations from the Zacks Database, 1985 to 1996

The number of listed firms includes all firms listed on the CRSP NYSE/AMEX/Nasdaq stock return file, by year. The number of covered firms is the number of firms with at least one valid recommendation in the Zacks database, by year. The number of covered firms is also expressed as the percent of the number of listed firms. The market capitalization of covered firms as a percent of the total market capitalization is the average daily ratio between the sum of the market capitalizations of all covered firms and the market value of all securities used in the CRSP daily value-weighted indices. The mean and median number of analysts issuing recommendations for each covered firm is shown, as is the mean and median number of firms covered by each analyst in the database, by year. This is followed by the number of brokerage houses and number of analysts with at least one recommendation during the year. The last column is the average of all analyst recommendations in the database for the year.

			Covered Firms		Analysts per Covered Firm		Covered Firms per Analyst				
Year (1)	No. of Listed Firms (2)	No. of Covered Firms (3)	As a % of Listed Firms (4)	Market Cap. As % of Market (5)	Mean (6)	Median (7)	Mean (8)	Median (9)	No. of Brokers (10)	No. of Analysts (11)	Average Rating (12)
1985	6,826	1,841	27.0	68.8	2.66	2	10	7	26	492	2.52
1986	7,281	2,989	41.1	85.3	4.25	3	13	10	61	960	2.37
1987	7,575	3,163	41.8	89.0	4.53	3	13	10	74	1,080	2.28
1988	7,573	3,226	42.6	90.5	4.75	3	13	10	96	1,171	2.32
1989	7,304	3,066	42.0	91.2	4.15	3	12	9	95	1,032	2.35
1990	7,138	3,105	43.5	92.3	4.50	3	13	10	98	1,082	2.34
1991	7,171	3,201	44.6	93.0	5.18	3	13	11	120	1,270	2.36
1992	7,459	3,546	47.5	93.8	5.09	3	12	10	131	1,452	2.23
1993	7,964	4,097	51.4	93.5	5.50	3	13	11	151	1,700	2.22
1994	8,494	4,611	54.3	93.9	5.61	3	13	11	169	2,007	2.09
1995	8,857	5,129	57.9	94.6	5.37	3	13	11	188	2,144	2.11
1996	9,408	5,628	59.8	95.6	5.27	3	13	11	195	2,367	2.04
Average All Years	7,754	3,634	46.1	90.1	4.74	3	13	10	117	1,396	2.27

Table I: Descriptive statistics on Zacks analyst recommendations.

Table II

Transition Matrix of Analyst Recommendations (Number, Median Calendar Days), 1985 to 1996

This table shows the number and the median calendar days between changes in or reiterations of recommendations. The first row reports all changes from a recommendation of 1 ("strong buy") to 1, 2 ("buy"), 3 ("hold"), 4 ("sell"), 5 ("strong sell") or discontinuation of coverage, and the total across the columns. The sixth and seventh rows identify recommendations for firms that were previously dropped from coverage and for firms for which coverage was initiated in the database. Fractional recommendations are rounded to the nearest whole value.

	To Recommendation of:						
From Recommendation of:	1	2	3	4	5	Dropped	Total
1	34,939	15,269	16,887	538	805	9,802	78,240
	293	109	128	140	135	121	
2	14,010	21,936	17,581	1,349	468	8,177	63,521
	95	299	115	106	111	121	
3	12,945	14,492	52,813	3,971	2,958	15,332	102,511
	113	112	291	114	116	123	
4	480	1,180	3,913	2,936	668	1,097	10,274
	132	103	98	245	98	135	
5	396	316	2,739	439	1,409	1,143	6,442
	95	105	94	90	301	99	
Dropped	4,951	3,507	5,999	546	400	5,013	20,416
	73	65	92	102	110	59	
First Zacks recommendation	26,053	19,817	24,458	2,392	1,531	5,965	80,216
Total	93,774	76,517	124,390	12,171	8,239	46,529	361,620
% of total	25.9	21.2	34.4	3.4	2.3	12.9	
% of non-drops	29.8	24.3	39.5	3.9	2.6		

Table II: Transition matrix of analyst recommendations.

6.3 Table III: announcement-window returns around recommendation changes

Read:

- Table III reports three-day market-adjusted returns around changes, reiterations, initial recommendations, and coverage drops.
- Upgrades are associated with positive announcement-window returns.
- Downgrades are associated with negative announcement-window returns.
- For example, a move from hold to strong buy has a positive return of 1.488%, while a move from strong buy to hold has a negative return of -2.192%.
- First Zacks recommendations are positive when the initial rating is a buy and negative when the initial rating is a hold or sell.

Economic message: Individual recommendation changes contain information at announcement. The later portfolio tests ask whether that information continues to predict returns after the initial announcement reaction.

Table III

Three-day Percentage Market-adjusted Returns Associated with Announcements of Changes in and Reiterations of Analyst Recommendations, 1985 to 1996

This table shows the percentage market-adjusted returns measured for the day before, the day of, and the day following changes in and reiterations of analyst recommendations. For example, the first row reports the returns associated with all changes from a recommendation of 1 (strong buy) to 1, 2 (buy), 3 (hold), 4 (sell), 5 (strong sell), or discontinuation of coverage. Returns are measured as the three-day buy and hold return less the return on a value-weighted NYSE/AMEX/Nasdaq index. The sixth and seventh rows show the returns associated with recommendations for firms that were previously dropped from coverage, and for firms for which coverage was initiated, respectively. Fractional recommendations are rounded to the nearest whole value. t-statistics, estimated using cross-sectional standard errors, are shown below the returns. Each t-statistic pertains to the hypothesis that the mean size-adjusted abnormal return is zero. (The number of observations in each cell is shown in Table II.)

From Recommendation of:	To Recommendation of: 1	2	3	4	5	Dropped
1	0.177	−0.889	−2.192	−1.305	−3.021	−0.020
	7.525	−17.448	−32.841	−4.129	−6.792	−0.364
2	1.059	0.114	−1.415	−0.638	−0.999	0.115
	21.565	3.809	−25.876	−3.154	−2.187	2.135
3	1.488	1.066	0.015	−1.054	−0.976	0.112
	27.895	22.877	0.788	−10.195	−5.926	2.630
4	0.723	0.610	0.610	−0.130	−0.336	0.393
	3.388	4.105	6.908	−1.399	−1.226	2.347
5	0.607	1.296	0.400	−0.283	−0.005	0.207
	2.113	4.384	3.487	−0.964	−0.032	0.999
Dropped	0.637	0.301	0.051	−1.168	−0.474	
	8.586	3.533	−0.810	−4.728	−1.463	
First Zacks recommendation	1.093	0.479	−0.149	−0.209	−0.650	
	29.445	13.150	−4.736	−2.135	−4.384	

Table III: Three-day market-adjusted returns around analyst recommendation events.

6.4 Tables IV and V: portfolio characteristics and baseline abnormal returns

Read Table IV:

- Table IV reports the composition and factor exposures of the recommendation portfolios.
- Portfolio 1 has an average of 760 stocks per month, while Portfolio 5 has only 258, reflecting the scarcity of unfavorable consensus ratings.
- Annual turnover is extremely high for the five recommendation portfolios, ranging from 433% to 478%.
- Portfolio 1 is tilted toward small growth stocks with relatively high market risk.
- Portfolio 5 is tilted toward small value stocks that have performed poorly in the recent past.

Read Table V:

- Table V is the baseline return table under daily rebalancing and immediate end-of-day reaction.
- Portfolio 1 earns a mean monthly market-adjusted return of 0.351%; Portfolio 5 earns -0.667%.
- The long-short Portfolio 1 minus Portfolio 5 strategy earns about 1.018% per month on a market-adjusted basis.
- The long-short abnormal gross return remains large after controls: 0.800% per month under the CAPM, 0.989% under Fama-French, and 0.753% under the four-characteristic model.

- After transaction costs, the active strategies no longer deliver reliably positive abnormal net annual returns.

Economic message: Analyst consensus recommendations strongly forecast gross returns. However, the same trading intensity that captures the signal creates large transaction costs.

Table IV

Descriptive Characteristics for Portfolios Formed on the Basis of Analyst Recommendations, 1986 to 1996

'his table presents descriptive statistics for several portfolios. The first five portfolios are based on the daily average analyst recommendation. 'ortfolios 1–5 include stocks with average daily recommendations of (1–1.5], (1.5–2], (2–2.5], (2.5–3] and greater than 3, respectively. The lifference between returns for portfolios 1 and 5 is shown next. The "All Covered" portfolio is the set of all stocks in portfolios 1–5, and the Neglected" portfolio consists of all stocks on the daily CRSP returns file with no Zacks recommendations for a sample day. The final line shows he difference between returns for the All Covered and Neglected stocks. The average monthly number of firms in each portfolio, the mean ıumber of analysts per firm per day in that portfolio, the average rating, and the percent of total market capitalization represented by the firms n the portfolio are shown. Annual turnover is calculated as the average percentage of the portfolio's holdings as of the close of one day's trading hat has been sold as of the close of trading on the next trading day, multiplied by the number of trading days in the year. The coefficient stimates are those from a time-series regression of the portfolio returns ($R_p - R_f$) on the market excess return ($R_m - R_f$), a zero-investment ize portfolio (SMB), a zero-investment book-to-market portfolio (HML), and a zero-investment price momentum portfolio (PMOM). t-statistics ppear below the coefficient estimates. Each t-statistic pertains to the null hypothesis that the associated coefficient is zero, except for the

Table IV, part 1: description of recommendation-portfolio characteristics.

Portfolio (1)	Monthly Avg. No. of Firms (min, max) (2)	No. of Analysts (3)	Average Rating (4)	% of Market Cap (5)	% Annual Turnover (6)	Coefficient Estimates for the Four-Characteristic Model: $R_m - R_f$ (7)	SMB (8)	HML (9)	PMOM (10)	Adjusted R^2 (11)
(most favorable)	760	2.35	1.24	8.5	458	1.055	0.214	−0.313	0.010	94.0
	(189, 1759)					**1.881**	**4.304**	**−5.408**	0.215	
	810	3.61	1.85	29.7	433	1.030	−0.020	−0.155	0.025	98.2
	(391, 1396)					**2.064**	−0.794	**−5.377**	1.049	
	646	4.93	2.29	34.2	459	0.988	−0.060	0.070	0.056	98.6
	(237, 948)					−1.055	**−3.110**	**3.118**	**3.023**	
	804	3.21	2.80	17.6	478	0.958	0.017	0.232	−0.022	95.8
	(522, 1046)					**−2.267**	0.538	**6.221**	−0.737	
(least favorable)	211	3.58	3.52	3.0	465	0.960	0.260	0.279	−0.293	88.3
	(115, 317)					−1.204	**4.556**	**4.213**	**−5.407**	
1–P5	971	NA	NA	NA	923	0.095	−0.046	−0.592	0.303	47.2
	(375, 1876)					**1.980**	−0.552	**−6.221**	**3.893**	
ll covered	3231	3.22	2.21	92.1	12	0.994	0.001	−0.004	0.015	99.9
	(1554, 5146)					**−1.698**	0.119	−0.607	**2.682**	
'eglected	3932	NA	NA	9.7	70	0.934	0.402	0.276	−0.024	94.9
	(3537, 4705)					**−3.120**	**11.161**	**6.554**	−0.698	
ll covered − neglected	7163	NA	NA	NA	82	0.060	−0.402	−0.280	0.039	60.9
	(6259, 8781)					**2.760**	**−10.744**	**−6.441**	1.095	

Table IV, part 2: portfolio characteristics, turnover, and factor exposures.

Table V

Percentage Monthly Returns Earned by Portfolios Formed on the Basis of Analyst Recommendations, 1986 to 1996

ıis table presents percentage monthly returns earned by portfolios formed according to average analyst recommendation. Raw returns are the ean percentage monthly returns earned by each portfolio. Market-adjusted returns are the mean raw returns less the return on a value eighted NYSE/AMEX/Nasdaq index. The CAPM intercept is the estimated intercept from a time-series regression of the portfolio return $_p - R_f$) on the market excess return ($R_m - R_f$). The intercept for the Fama-French three-factor model is the estimated intercept from a me-series regression of the portfolio return on the market excess return ($R_m - R_f$), a zero-investment size portfolio (SMB), and a zero-vestment book-to-market portfolio (HML). The four-characteristic intercept is estimated by adding a zero-investment price momentum port-lio (PMOM) as an independent variable. Annual turnover is calculated as the average percentage of the portfolio's holdings as of the close of ıe day's trading that has been sold as of the close of trading on the next trading day, multiplied by the number of trading days in the year. The ɪt annual return assumes that portfolios 1 and 2 are purchased, and 3, 4, and 5 are sold short. It is found by multiplying the absolute ılue of the gross monthly return by 12 and subtracting the annual turnover multiplied by the round-trip cost of a trade. This cost is estimated 1.31 percent for all portfolios except that of the neglected stocks, where the estimated cost is 4.12 percent. Each t-statistic pertains to the

Table V, part 1: description of baseline portfolio return tests.

Portfolio (1)	Mean Raw Return (2)	Mean Market-adjusted Return (3)	Intercept from: CAPM (4)	Fama-French (5)	Four-characteristic (6)	% Annual Turnover (7)	Net Annual Return from: CAPM (8)	Fama-French (9)	Four-characteristic (10)
(most favorable)	1.576	0.351	0.201	0.352	0.344	458	−3.586	−1.772	−1.867
		2.472	1.475	**3.167**	**2.930**				
	1.495	0.270	0.184	0.229	0.210	433	−3.470	−2.919	−3.149
		3.999	**2.976**	**4.140**	**3.605**				
	1.263	0.038	0.029	−0.006	−0.049	459	−5.663	−5.942	−5.422
		0.763	0.592	−0.132	−1.076				
	1.121	−0.103	−0.053	−0.124	−0.107	478	−5.629	−4.773	−4.982
		−1.180	−0.640	**−1.729**	−1.409				
(least favorable)	0.558	−0.667	−0.599	−0.637	−0.409	465	1.099	1.552	−1.179
		−3.908	**−3.502**	**−4.513**	**−3.044**				
1–P5	1.018	1.018	0.800	0.989	0.753	923	−2.491	−0.223	−3.055
	4.160	**4.160**	**3.495**	**5.113**	**3.900**				
ll covered	1.306	0.081	0.053	0.055	0.043	12	0.479	0.502	0.363
		6.432	**3.994**	**4.102**	**3.150**				
eglected	0.890	−0.334	−0.277	−0.273	−0.254	70	0.443	0.392	0.168
		−2.799	**−2.245**	**−3.371**	**−2.977**				
ll covered − neglected	0.416	0.416	0.330	0.328	0.298	82	0.922	0.894	0.531
	3.200	**3.200**	**2.606**	**3.910**	**3.374**				

Table V, part 2: gross monthly returns, turnover, and net annual returns.

6.5 Tables VI and VII: rebalancing frequency and delayed reaction

Read Table VI:

- Table VI studies weekly, semimonthly, and monthly rebalancing instead of daily rebalancing.
- Less frequent rebalancing sharply lowers turnover.
- The abnormal gross returns for Portfolio 1 become smaller and often statistically insignificant.
- Portfolio 5 continues to earn significantly negative abnormal gross returns across the alternative rebalancing frequencies.

Read Table VII:

- Table VII keeps daily rebalancing but assumes investors react to recommendation changes after one week, half a month, or one month.
- For Portfolio 1, delayed reaction reduces abnormal gross returns to near zero.
- For Portfolio 5, abnormal gross returns remain significantly negative even after delays.

- The results imply that buying the most favorable recommendations requires fast reaction, while the underperformance of unfavorable recommendations persists longer.

Economic message: Implementation timing is central. The positive side of the recommendation strategy decays quickly, while the negative side is more persistent.

Percentage Monthly Returns Earned by Portfolios Formed on the Basis of Analyst Recommendations, by Rebalancing Frequency, 1986 to 1996

his table presents percentage monthly returns earned by portfolios composed of the most favorably and least favorably ranked stocks, for ırious rebalancing periods. Panel A presents the returns based on a strategy of rebalancing the portfolios weekly, at the close of trading each londay. Panel B presents the returns to a strategy of rebalancing the portfolios semimonthly, at the close of trading on the 15th and last days the month. Panel C presents the returns to a strategy of rebalancing the portfolios monthly, at the close of trading on the last day of the month. aw returns are the mean percentage monthly returns earned by each portfolio. Market-adjusted returns are the mean raw returns less the eturn on a value weighted NYSE/AMEX/Nasdaq index. The CAPM intercept is the estimated intercept from a time-series regression of the ortfolio return $(R_p - R_f)$ on the market excess return $(R_m - R_f)$. The intercept for the Fama-French three-factor model is the estimated itercept from a time-series regression of the portfolio return on the market excess return $(R_m - R_f)$, a zero-investment size portfolio (SMB), and zero-investment book-to-market portfolio (HML). The four-characteristic intercept is estimated by adding a zero-investment price momentum ortfolio (PMOM) as an independent variable. Annual turnover is calculated as the average percentage of the portfolio's holdings as of the close one day's trading that has been sold as of the close of trading on the next trading day, multiplied by the number of trading days in the year. he net annual return assumes that portfolio 1 is purchased and 5 is sold short. It is found by multiplying the absolute value of the gross ıonthly return by 12 and subtracting the annual turnover multiplied by the round trip cost of a trade. This cost is estimated at 1.31 percent for l portfolios. Each t-statistic pertains to the null hypothesis that the associated return is zero. The t-statistics for returns that are significant a level of 10 percent or better are shown in bold.

Portfolio	Mean Raw Return	Mean Market-adjusted Return	Intercept from: CAPM	Intercept from: Fama-French	Intercept from: Four-characteristic	% Annual Turnover	Net Annual Return from: CAPM	Net Annual Return from: Fama-French	Net Annual Return from: Four-characteristic
Panel A: Weekly Rebalancing									
1 (most favorable)	1.483	0.258	0.079	0.233	0.182	395.2	−4.232	−2.386	−2.996
		1.699	0.560	**1.997**	1.487				
5 (least favorable)	0.687	−0.538	−0.479	−0.526	−0.329	377.5	0.805	1.370	−0.995
		−3.168	**−2.794**	**−3.791**	**−2.427**				
1–P5	0.796	0.796	0.558	0.759	0.511	772.7	−3.427	−1.016	−3.990
	3.072	**3.072**	**2.316**	**3.770**	**2.544**				
Panel B: Semimonthly Rebalancing									
1 (most favorable)	1.479	0.255	0.082	0.234	0.212	346.4	−3.555	−1.727	−1.995
		1.708	0.591	**2.037**	**1.745**				
5 (least favorable)	0.609	−0.615	−0.555	−0.599	−0.368	349.3	2.089	2.612	−0.157
		−3.606	**−3.233**	**−4.175**	**−2.692**				
1–P5	0.870	0.870	0.637	0.833	0.580	695.7	−1.465	0.884	−2.152
	3.421	**3.421**	**2.694**	**4.227**	**2.966**				
Panel C: Monthly Rebalancing									
1 (most favorable)	1.417	0.192	0.031	0.188	0.181	273.5	−3.205	−1.326	−1.412
		1.278	0.221	1.591	1.448				
5 (least favorable)	0.627	−0.598	−0.535	−0.577	−0.378	293.5	2.570	3.080	0.692
		−3.621	**−3.223**	**−4.240**	**−2.857**				
1–P5	0.790	0.790	0.566	0.765	0.559	567	−0.635	1.754	−0.720
	3.197	**3.197**	**2.457**	**4.020**	**2.904**				

Table VI, part 1: description of rebalancing-frequency tests.

Table VI, part 2: weekly, semimonthly, and monthly rebalancing results.

Table VII

Percentage Monthly Returns Earned by Portfolios Formed on the Basis of Analyst Recommendations, by Delay in Investment, 1986 to 1996

This table presents percentage monthly returns earned by portfolios composed of the most favorable and least favorable ranked stocks, where investment is delayed beyond the close of trading on the date the average recommendation changes. Panel A presents the results for a one-week delay, Panel B for a half-month delay, and Panel C for a one-month delay. Raw returns are the mean percentage monthly returns earned by each portfolio. Market-adjusted returns are the mean raw returns less the return on a value weighted NYSE/AMEX/Nasdaq index. The CAPM intercept is the estimated intercept from a time-series regression of the portfolio return ($R_p - R_f$) on the market excess return ($R_m - R_f$). The intercept for the Fama-French three-factor model is the estimated intercept from a time-series regression of the portfolio return on the market excess return ($R_m - R_f$), a zero-investment size portfolio (SMB), and a zero-investment book-to-market portfolio (HML). The four-characteristic intercept is estimated by adding a zero-investment momentum portfolio (PMOM) as an independent variable. Each t-statistic pertains to the null hypothesis that the associated return is zero. The t-statistics for returns that are significant at a level of 10 percent or better are shown in bold.

			Intercept from		
Portfolio (1)	Mean Raw Return (2)	Mean Market-adjusted Return (3)	CAPM (4)	Fama-French (5)	Four-characteristic (6)
Panel A: One-week Delay					
P1 (most favorable)	1.422	0.198	0.025	0.174	0.158
		1.267	0.170	1.394	1.198
P5 (least favorable)	0.699	−0.526	−0.467	−0.518	−0.335
		−3.118	**−2.750**	**−3.767**	**−2.468**
P1–P5	0.723	0.723	0.492	0.692	0.493
	2.838	**2.838**	**2.073**	**3.450**	**2.412**
Panel B: Semimonthly Delay					
P1 (most favorable)	1.408	0.181	0.034	0.177	0.181
		1.273	0.249	1.524	1.478
P5 (least favorable)	0.809	−0.418	−0.359	−0.403	−0.223
		−2.541	**−2.170**	**−3.008**	−1.693
P1–P5	0.599	0.599	0.393	0.580	0.404
	2.467	**2.467**	1.716	**3.015**	**2.054**
Panel C: One-month Delay					
P1 (most favorable)	1.283	0.056	−0.081	0.077	0.084
		0.386	−0.566	0.659	0.681
P5 (least favorable)	0.854	−0.373	−0.331	−0.388	−0.229
		−2.329	**−2.032**	**−3.234**	−1.940
P1–P5	0.429	0.429	0.251	0.465	0.313
	1.797	1.797	1.090	**2.539**	1.662

Table VII: Portfolio returns when investors react with delays.

6.6 Table VIII: size partitions

Read:

- Table VIII reports returns separately for small, medium, and large firms.
- The small-firm recommendation spread is strongest: the four-characteristic monthly gross return is 0.575% for Portfolio 1 and -0.926% for Portfolio 5.
- For medium firms, the corresponding spread remains positive and statistically meaningful.
- For large firms, the Portfolio 1 minus Portfolio 5 spread is much smaller and statistically insignificant.
- Once size-specific trading costs are applied, net annual returns are negative for the reported strategies.

Economic message: The recommendation signal is concentrated where informational frictions are most plausible - small stocks. But those are also the stocks where transaction costs are largest.

Table VIII

Percentage Gross Monthly and Net Annual Returns Earned by Portfolios Formed on the Basis of Analyst Recommendations and Size, 1986 to 1996

'his table presents the percentage gross monthly and net annual returns earned by portfolios formed by average analyst recommendations and firm size. 'he large (small) firm sample, B (S), includes firms with market capitalizations in the top (bottom) 30 percent of NYSE firms. The medium-sized firm ample, M, includes firms with market capitalizations between the 30th and 70th percentiles of NYSE firms. Raw returns are the mean percentage ıonthly returns earned by each portfolio. (Underneath each of the raw returns for portfolios 1–5 is the average monthly number of firms in that portfolio.) Iarket-adjusted returns are the mean raw returns less the return on a value weighted NYSE/AMEX/Nasdaq index. The gross monthly return for the ɔur-characteristic model is the estimated intercept from a time-series regression of the portfolio excess return on the market excess return ($R_m - R_f$), a ero-investment size portfolio (SMB), a zero-investment book-to-market portfolio (HML), and a zero-investment price momentum portfolio (PMOM). .nnual turnover is calculated as the average percentage of the portfolio's holdings as of the close of one day's trading that has been sold as of the close f trading on the next trading day, multiplied by the number of trading days in the year. The net annual return assumes that portfolios 1 and 2 are urchased, and 3, 4, and 5 are sold short. It is found by multiplying the absolute value of the gross monthly return by 12 and subtracting the annual ırnover multiplied by the round trip cost of a trade. This cost is estimated at 0.727 percent for big firms, 1.94 percent for medium-sized firms, and .12 percent for small firms. Each t-statistic pertains to the null hypothesis that the associated return is zero. The t-statistics for returns that are ignificant at a level of 10 percent or better are shown in bold.

	Mean Raw Return			Mean Market-Adjusted Return			Gross Monthly Return from Four-Characteristic Model			% Annual Turnover			Net Annual Return from Four-Characteristic Model		
	S	M	B	S	M	B	S	M	B	S	M	B	S	M	B
Portfolio	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
(most favorable)	1.800	1.654	1.468	0.575	0.430	0.244	0.575	0.387	0.251	265	409	618	−4.014	−3.285	−1.479
	560	114	17	**2.283**	**2.253**	1.213	**5.615**	**2.715**	1.293						
	1.478	1.589	1.482	0.253	0.365	0.257	0.327	0.226	0.212	384	450	462	−11.895	−6.021	−0.819
	475	216	95	1.155	**2.557**	**2.843**	**3.602**	**2.314**	**2.730**						
	1.253	1.309	1.270	0.029	0.084	0.045	−0.004	−0.027	−0.022	497	458	487	−20.425	−8.558	−3.272
	261	238	141	0.142	0.837	0.561	−0.041	−0.347	−0.366						
	0.796	1.061	1.200	−0.429	−0.164	−0.025	−0.275	−0.169	−0.032	309	406	575	−9.426	−5.843	−3.792
	523	200	72	**−2.363**	**−1.585**	−0.193	**−3.717**	**−1.932**	−0.305						
(least favorable)	0.040	0.675	0.716	−1.184	−0.550	−0.508	−0.926	−0.596	−0.017	357	403	638	−3.594	−0.661	−4.434
	139	59	12	**−4.234**	**−2.960**	**−1.818**	**−5.057**	**−3.695**	−0.066						
'1–P5	1.759	0.979	0.752	1.759	0.979	0.752	1.502	0.984	0.268	622	812	1256	−7.608	−3.946	−5.913
	6.893	**4.025**	**2.040**	**6.893**	**4.025**	**2.040**	**7.302**	**4.516**	0.799						

Table VIII: Gross monthly and net annual returns by recommendation portfolio and firm size.

7 Short summary

- The paper asks whether investors can profit from public analyst consensus recommendations.
- It uses Zacks recommendations from 1985 to 1996 and CRSP returns.
- Ratings are averaged across analysts to form a consensus recommendation for each firm.
- Stocks are sorted daily into five value-weighted portfolios from most favorable to least favorable.
- The most favorably recommended stocks earn much higher raw and risk-adjusted gross returns than the least favorably recommended stocks.
- The long-short gross return is large under CAPM, Fama-French, and momentum-augmented models.
- Less frequent rebalancing and delayed reaction reduce the positive returns of the most favorable portfolio.
- The negative returns of the least favorable portfolio are more persistent.
- The strategies require turnover often above 400% per year.
- After realistic trading costs, the abnormal net returns are not reliably positive.
- Predictability is concentrated in small firms; it is weak among large firms.
- Analyst recommendations are informative, but exploiting them as an active trading strategy is difficult after implementation costs.

8 Conclusion

8.1 Core conclusion

Barber, Lehavy, McNichols, and Trueman show that analyst consensus recommendations contain information about future stock returns. Stocks with the most favorable recommendations outperform stocks with the least favorable recommendations on a gross basis, even after controlling for market risk, size, book-to-market, and momentum.

8.2 Economic intuition

Analysts may help process firm-specific information, and prices may not fully incorporate recommendation changes immediately. Highly recommended stocks tend to continue performing well, while poorly recommended stocks continue underperforming. This generates return predictability in the short and medium run.

8.3 Implementation friction

The profitable-looking gross strategy is not free to implement. Because portfolios must respond quickly to frequent recommendation changes, the turnover is very high. Once transaction costs are deducted, the net abnormal return from the main active strategies is not reliably above zero.

8.4 Market efficiency interpretation

The paper presents a nuanced view of market efficiency. The gross-return evidence rejects the idea that recommendations are instantly and fully impounded in prices. The net-return evidence is more consistent with a world in which predictability exists but is difficult to arbitrage away because trading costs are large.

8.5 Implications

The findings imply that analyst recommendations can be useful for investors' trading and portfolio decisions, especially for avoiding poorly rated stocks. But the paper does not support the claim that simple, high-turnover trading strategies based on consensus recommendations reliably generate positive abnormal net returns.

8.6 Limitations

The database omits some recommendations available to brokerage clients, and the results may depend on the timing of recommendation reporting. The analysis is also sample-period specific, covering 1986 to 1996 returns. Finally, transaction-cost estimates are approximate and may vary across investor types and execution methods.

8.7 Takeaway

The enduring takeaway is that analyst recommendations matter, but profits from them are partly competed away by the cost of acting on them. Investors can observe economically meaningful gross return patterns, yet capturing those patterns requires speed, trading intensity, and low costs that many investors do not have.